

Human Research Ethics Application

Application Management Information

Application ID: [REDACTED]

Created date: 11/05/2023

Originating Application ID:

**This is the earliest application from which this application ([REDACTED]) was copied.*

Parent Application ID:

**This is the immediate predecessor from which this application ([REDACTED]) was copied.*

Version Number: 1

Application submitted to: [REDACTED]; [REDACTED] Human Research Ethics Committee.

The applicant has requested that this ethics application be considered under the Greater than low risk review pathway.

Section 1 – Core Information

Pre-application conditions

The applicant/s have acknowledged that:

1. The HREA has been designed for ethics review of human research, as defined in the [National Statement](#).
2. Adequate resources must be available to conduct this research project.
3. All relevant institutional policies pertaining to the conduct of this research project should be considered and adhered to.
4. Research activities must not commence until ethics approval (and site authorisation, if appropriate) has been provided.

Project Overview

Q1.1 Project Title:

How Autistic Preschool Children Undertake Novel Word Learning from Social Robots

Q1.2 Summary of the research project:

The present study aims to investigate how autistic children learn novel words from social robots and how word learning may be affected by engagement with the robot and in the task. Thirty-eight autistic children, aged 3 to 5 years old will be recruited from an early intervention facility, the AEIOU Foundation. Participants will listen to an e-book read by a social robot that contains four novel words with corresponding referent pictures, and then asked a series of closed ended questions with visuals. Engagement behaviour will be video recorded and coded for analysis. Existing data from a previous study (ethics approval number, [REDACTED] no.: 2022/405) for thirty-eight neurotypical 3 to 5 year old children, previously collected from day-care centres with identical methodology, will be used for comparison. Based on previous research, social robots are expected to support learning of novel words and engagement of autistic children.

Q1.3 Which category/ies of research best describes the project?

Psychology; Autism; Social Robots; Learning

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Q1.4 In what environments will the research be conducted?

Community centre(s)

Q1.5 What organisation/entity has overall responsibility for this project?

██████████ has overall responsibility for this project.

Q1.6 Describe how this research project is currently, or will be, funded.

Any research costs will be covered within ██████████'s Honours funds (Total funds available = \$150).

Q1.7 Anticipated starting date of the research project:

As soon as ethics and any other relevant approvals have been provided.

Q1.8 Anticipated duration of the research project:

7 Months

Project Team

Name: [REDACTED]

Q1.9.4 Email Address:

[REDACTED]

Q1.9.5 Is this person the contact person for this application?

No

Q1.9.6 Is this person a student on this project?

No

Q1.9.7 Institutional affiliation and position:

Honours student (Bachelor of Psychology), [REDACTED], [REDACTED].

Q1.9.8 Staff ID (optional):

Q1.9.9 ORCID Identifier (optional):

Q1.9.10 Position on the research project:

Investigator/Researcher

Q1.9.11 Does this person have authorisation to sign the application on behalf of all members of the research team?

Yes

Q1.9.12 Research activities [REDACTED] will be responsible for:

[REDACTED] is an honours student who will conduct all experiments with participants, under the supervision of [REDACTED] and [REDACTED]. She will be principally responsible for participation recruitment, obtaining consent, data collection, coding and writing up the results, and drafting and revising the broader thesis. This study will be submitted as a partial requirement for the degree of Bachelor of Psychology (honours) in the [REDACTED] at [REDACTED], Gold Coast campus.

Q1.9.13 Expertise relevant to the research activity:

[REDACTED] has completed a Bachelor of Psychology and is a current psychology honours student. She has conducted research assistance and data collection with [REDACTED] and has experience as a disability support worker, primarily with autistic children.

Name: [REDACTED]

Q1.9.4 Email Address:

[REDACTED]

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Q1.9.5 Is this person the contact person for this application?

No

Q1.9.6 Is this person a student on this project?

No

Q1.9.7 Institutional affiliation and position:

██████████ Postdoctoral Research Fellow Awardee

Q1.9.8 Staff ID (optional):

Q1.9.9 ORCID Identifier (optional):

Q1.9.10 Position on the research project:

Principal Investigator

Q1.9.11 Does this person have authorisation to sign the application on behalf of all members of the research team?

Yes

Q1.9.12 Research activities ██████████ will be responsible for:

██████████ is ██████████'s principal supervisor and will be primarily responsible for supervision throughout the research process, including assisting with recruitment, providing clinical guidance and training for data collection, and editing and revising the write-up of the results and broader thesis.

Q1.9.13 Expertise relevant to the research activity:

██████████ is an autistic early career researcher and has been conducting and supervising child-robot interaction research with children between 1 and 11 years of age for almost a decade. She has experience of direct relevance to this study including data collection within this projects specific age group and methodology to provide the supports needed for ██████████ to complete her project.

Name: ██████████

Q1.9.4 Email Address:

Q1.9.5 Is this person the contact person for this application?

Yes

Q1.9.5.1 Email Address:

**Q1.9.5.2 Telephone
Number:**

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Q1.9.5.3 Mailing Address [REDACTED], G40 7.35, Gold Coast Campus

Q1.9.6 Is this person a student on this project?

No

Q1.9.7 Institutional affiliation and position:

Senior Lecturer, [REDACTED], [REDACTED]

Q1.9.8 Staff ID (optional):

Q1.9.9 ORCID Identifier (optional):

Q1.9.10 Position on the research project:

Associate/Assistant/Sub-/Co- Investigator/Researcher

Q1.9.11 Does this person have authorisation to sign the application on behalf of all members of the research team?

Yes

Q1.9.12 Research activities [REDACTED] will be responsible for:

[REDACTED] is [REDACTED]'s associate supervisor and will be responsible for providing clinical guidance and training for data collection and editing and revising the write-up of the results and broader thesis.

Q1.9.13 Expertise relevant to the research activity:

[REDACTED] has two decades of experience working with this population and is a clinical psychologist. She has experience of direct relevance to this study including data collection and working with this specific clinical population to provide the supports needed for [REDACTED] to complete her project and ensure data collection and methods are sensitive and supportive of this population.

Name: [REDACTED]

Q1.9.4 Email Address:

Q1.9.5 Is this person the contact person for this application?

No

Q1.9.6 Is this person a student on this project?

No

Q1.9.7 Institutional affiliation and position:

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Research Assistant, [REDACTED], [REDACTED]

Q1.9.8 Staff ID (optional):

Q1.9.9 ORCID Identifier (optional):

Q1.9.10 Position on the research project:

Associate/Assistant/Sub-/Co- Investigator/Researcher

Q1.9.11 Does this person have authorisation to sign the application on behalf of all members of the research team?

Yes

Q1.9.12 Research activities [REDACTED] will be responsible for:

[REDACTED] is a research assistant to this project, providing her expertise around the methodology and data collection as she has previous experience with the experimental procedure.

Q1.9.13 Expertise relevant to the research activity:

[REDACTED] conducted the original neurotypical sample of this experiment and thus has experience conducting this specific research study in childcare centres with over 80 children. She has also completed her Bachelor of Psychology (honours) and is a research assistant for [REDACTED].

Disclosure of interests

Q1.10 Do any members of the research team (including persons not listed in this application), have any financial or non-financial interests related to this research?

No

Restrictions

Q1.11 Are there any restrictions or limits on publication of data or dissemination of research outcomes of this project?

No

Evaluations

Q1.12 Has the scientific or academic merit of the research project been evaluated?

Yes

Q1.12.1 What was the review process and what was the outcome?

The project proposal was reviewed via an internal process conducted by [REDACTED] through review of the project proposal. The proposal was approved and deemed satisfactory on the 10th of May 2023 by Darren Wishart and Matthew Gullo.

The procedure for children assessment has also obtained previous ethical approval for neurotypical children through [REDACTED], [REDACTED] no.: 2022/405.

AEIOU Foundation has further provided conditional gatekeeper approval for the current study, (approval number: AEIOU 2023-2) subject to ethical clearance. See attached ARIC Decision Letter. This was sought to ensure the proposed design and approach was appropriate to the organisation and autistic children to review the scientific merit and appropriateness of the proposed project.

Q 1.12.2 Attach evidence of the outcome of the scientific or academic review process.

(optional)

[REDACTED] Honours Proposal passed letter 2023.pdf

Q1.13 Has this research project had prior ethics review?

No

Q1.14 Will any further or additional specialised review of this application be sought?

No

Setting of research

Q1.15 Will this project be conducted at multiple sites?

Yes

Q1.16 Will separate institutional approvals or authorisations be required prior to commencing research at each site?

Yes

Section 2 – Research Details and Participants

Q1.17 The following research methods will be used in the research project:

Research Method	Status
Action research	X
Biospecimen analysis research	X
Data linkage research	X
Ethnographic research	X
Epidemiological research	X
Interventional/Clinical Trials research	✓
Observational research	X
Survey/Interview/Focus Group research	X
Textual analysis research	X
None of the above	X

Q1.18 The research will be conducted with the following:

Participation	Status
Human beings (via active participation), including their associated biospecimens and/or data.	✓
Human biospecimens only	X
Data associated with human beings only (i.e. as the primary object of research)	X

Q1.19 The research will involve the following participants:

Participants	Status
Women who are pregnant and the human fetus	X
Children and young people	✓
People highly dependent on medical care who may be unable to give consent	X
People with a cognitive impairment, intellectual disability or mental illness	✓
People in dependent or unequal relationships	X
People who may be involved in illegal activities	X
People in other countries	X
Aboriginal and Torres Strait Islander peoples	X

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Method Specific Questions

Interventional/Clinical Trials Research

M6.1 Briefly describe the intervention/s that you will be using.

The current research will explore the potential utility of social robots to facilitate word learning with 3-5-year old autistic children. Children, recruited from AEIOU centres, will play with a social robot (NAO, Softbank Robotics) and engage in a book reading activity with the social robot. The whole task takes 10 minutes in total per child. The design of this research is based off a series of successful social robot mediated word learning studies with neurotypical children (NT), illustrating successful use of the methodologies with the NT 3- 5 year age range. The tasks require no expressive verbal ability (i.e. no talking) and instead explores children's understanding of the story and offers multiple methods for responding, such as pointing at pictures. Children will be presented with a 2-option task to measure their learning and we will explore whether word learning is better than chance (chance being whether children will be choosing answers at random or based off their knowledge garnered from the book read by the social robot.) This study will replicate a study conducted with NT children (██████ no.: 2022/405) enabling us to compare autistic children to NT children. In addition, to characterise children's abilities we will ask consent from parents to access existing measures of non-verbal ability, adaptive functioning, and autism characteristics which have already been collected by the AEIOU Foundation as part of their standard educational practice. Requesting access to this data is to minimise burden on children and their parents.

M6.2 Is your intervention related to the prevention, diagnosis, treatment or management of a health condition?

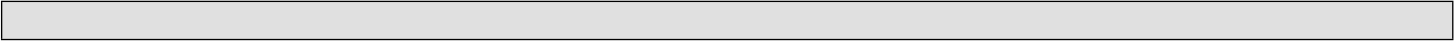
No

M6.3 With regard to your answers above, describe any ethical considerations related to your use of the intervention/s in this research project and your plans for addressing these issues.

Confidentiality has been considered with video footage taken of the children interacting with the social robot to code behavioural engagement. After coding is complete, the video footage will be stored on a secure hard drive and on the ████████ Research Data Management Server. Data will be retained in a deidentified form by the ████████ ████████. The research involves children 3 to 5 years old with autism spectrum disorder (ASD) which will be sourced from the AEIOU Foundation. Consent and assent have been considered and all children will have prior parental approval and consent and the parents and child will be informed about what is expected and will provide their assent verbally or via their behaviour. In the unlikely event of any signs of distress or discomfort, this will be taken as children withdrawing assent and participation discontinued.

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Participant Specific Questions

Children and young people

P2.1 How will the children or young people participate in this research?

Children will be instructed on how to complete a 'low-five' with NAO (social robot) and then asked to watch a 30-second video of a child playing with blocks with NAO. During the interaction, NAO will run through a pre-programmed script in which NAO introduces itself, invites the child to play with blocks, and asks the child questions before reading the child a short e-book (note: verbal responses are not required to robot questions/prompts). The story depicts a child deciding what to take for show and tell at school. The book will display the four novel words, WUG, LIF, POK and NEEM, three times each before proceeding to the test phase. The test phase will be presented in a two-response forced choice format where the child will be asked which target novel word matches the picture (the child's response may be either verbal or gestural). The participant will be directed to select their answer by either pointing or touching their response on an iPad. Following the task, NAO will say goodbye to the child and the child will be directed or assisted back to their program. The task will take a total of 10 minutes and the time will be arranged with AEIOU staff to minimise disruption to their routines, with verbal or visual supports used (as selected by their educators) to inform children of the change in routine and introduce the examiner to support an enjoyable experience. In the past, we have found children have typically enjoyed this procedure.

P2.2 Explain how the research is likely to advance knowledge about the health or welfare or other matters relevant to children or young people.

This research will help us to understand how engaging with and how effective social robots are as a learning tool for autistic children, building on previous research suggesting potential benefits for therapeutic outcomes such as social interaction skills. This research will explore the potential benefits of social robots in increasing engagement and learning of novel words which may inform potential interventions for aiding in a successful transition from an early intervention setting to a school setting and target early language learning in autistic preschool children.

P2.3 Explain whether children's or young people's participation is indispensable to the conduct of the research.

As the research is focused on the learning of autistic children, this group are indispensable to the conduct of this research, as direct assessment and measurement is required.

P2.4 How will you ensure that the children or young people's safety, emotional and psychological security and wellbeing are protected?

The procedure has been selected based on previous successful research and adaptations (prior introduction, use of visuals to let children know of the session) drawn from our clinical and research experience to ensure children are informed to the best of their ability of what is happening, are given opportunities to provide assent, and decline participation. Further, by using non-verbal response options, tasks are tailored to the communication challenges and strengths of this population. Assessments are non-invasive and students are able to stop or leave at any time. The types of activities that participants will be involved are play based to make it as enjoyable as possible for them.

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P2.5 How will you establish that participation in the research is not contrary to the best interests of the children or young people?

Parents will be provided with an information package which will provide details of the study and the assessments/procedure their child will undergo if they choose to participate. This information sheet will outline that participation is entirely voluntary and no consequences will be had for non-participation. Consent from one parent will be obtained in writing for each participant prior to any assessments being conducted. Assent will also be obtained by the participant prior to the experiment by the child showing assent in their behaviour (e.g., gesture, willingness to follow the examiner to the room) or verbally if they have verbal ability.

People with a cognitive impairment, intellectual disability or mental illness

P4.1 Who will the research involve?

People with a cognitive impairment
People with an intellectual disability

P4.2 How will the research project accommodate people with a cognitive impairment, an intellectual disability or a mental illness?

Parents will be provided with an information package which will provide details of the study and the assessments/procedure their child will undergo if they choose to participate. This information sheet will outline that participation is entirely voluntary and no consequences will be had for non-participation. Informed consent will be obtained from parents/caregivers prior to any assessments being conducted. The types of assessments being performed with participants will be specifically developed for children in this age group and will be tailored to individual capabilities. Assessments are non-invasive and children will be given a break when needed. Both [REDACTED] and [REDACTED] have extensive clinical experience in this area that will guide the approach to ensure the needs of this group will be accommodated. Further, to help minimise any distress of the participants, where children are using a visual schedule, a photograph of the robot will be placed on their schedule after discussion with room staff/centre managers. We will also provide a social story of the experiment for children prior to engaging in the study and they will be provided with a video to watch before the experiment of a child playing with the social robot. Verbal responses are not required to answer robot questions/prompts and the child is able to consent to participation either verbally or gesturally.

P4.3 How will the participants' degree of cognitive impairment, intellectual disability or mental illness be assessed?

We will request parent consent to access existing measures completed by the AEIOU Foundation to minimise burden on participants. These will include:

1. Autism Diagnostic Observation Schedule (ADOS) results will be used to confirm level of autism characteristics.
2. Mullen Scales of Early Learning (MSEL) scores will be used to evaluate the level of cognitive functioning of the child.
3. Vineland Adaptive Behaviour Scales (VABS) will be used to determine the adaptive behaviour of the child.

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P4.4 How will you assess whether participants' cognitive impairment, intellectual disability or mental illness increases their susceptibility to discomfort or distress?

Assessment will be undertaken in consultation with AEIOU staff and parents as well as information garnered from preliminary assessment (ADOS, MSEL, and VABS). We anticipate children will generally enjoy the play-based technology activity, however, in the rare chance any participants with autism dislike or feel discomfort with the robot or situation, they may stop or leave at any time. Furthermore, to minimise risk of any distress arising from the novelty of the situation or the change in routine, children will be offered to watch a video before agreeing to participate.

Recruitment Questions

Q2.1.1 Indicate how you will identify and recruit participants for your research, referencing any relevant sections of your Project Description/Protocol as appropriate.

Participants will consist of 38 autistic children aged 3 to 5 years old of any gender. This sample size is to match a currently completed cohort of neurotypical participants to enable comparisons. Participants will be recruited from AEIOU centres across Brisbane and Gold Coast who are currently attending programs at AEIOU. As the research requires participants to listen to and respond to the social robot, those with hearing and visual problems are unable to participate. Children will be included based on: 1) a clinical diagnosis of autism, 2) the child does not show risk of harm to themselves or others (as identified by AEIOU staff and/or parents) and 3) are willing to participate in a play-based environment. Parents will be required to complete a consent form for their child to participate and to provide consent to access existing child data. Recruitment will occur through advertising via paper forms initially handed directly to caregivers and via QR codes linked to redcap survey, followed by email, flyers, newsletters, and website distributed and displayed within the centres.

Q2.1.2 How will your recruitment strategy take account of the ethical considerations relevant to the specific people you are recruiting?

Based on previous clinical research, the profile and number of projected participants is appropriate to answer the research questions. Prospective parents of participants will be provided with an information sheet outlining all potential risks and benefits to their potential participation in the research to provide their informed consent. Letters will outline the description of the study, what is expected of the child, and how to participate in it. Parents will be required to complete a consent form for their child to participate and to provide consent to access existing child data to reduce burden on children and families. Participants will be welcome to contact members of the research team if they have any questions regarding the information sheet provided to them about the project. The decision to participate will be voluntary and participants will be free to decline or desist participation at any point. If prospective participants do not respond or indicate they do not wish to participate, their participation will respectively cease.

Consent Questions

Q2.2.1 Indicate by reference the relevant section/s of your Project Description/Protocol that address/es consent.

Parents of the child will be given an information sheet and informed consent form for both their child's participation and access to previous assessment data. This will be issued by honours student [REDACTED].

Q2.2.2 Will you be obtaining consent from some or all participants to participate in the research?

Yes for all participants

Q2.2.2.1 What is the scope of consent that you will be seeking?

Specific

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Q2.2.2.2 How will consent be obtained?

Written

Q2.2.2.3 Are you proposing to obtain consent using limited disclosure?

No

Q2.2.3 Are family members, authorised representatives or any others involved in the participants' decision to participate in the research?

Yes

Q2.2.3.1 Outline the family members, authorised representatives or others that will be involved in the participants' decision to participate in the research and the extent of their involvement.

The decision for child participation will be the responsibility of one parent/primary caregiver.

Q2.2.4 Will there be an opportunity to confirm or re-negotiate consent during the research project?

Yes

Q2.2.6 Describe any ethical considerations related to the approach to consent that you will be seeking and your strategies for addressing and managing these issues.

No ethical considerations related to this approach are anticipated.

Q2.2.7 Are you proposing to use an opt-out approach with respect to some or all of the participants?

No

Q2.2.8 Are you requesting a waiver of the requirement for consent with respect to some or all participants?

No

(Children and young people specific questions)

Q2.2.P2.1 From whom are you obtaining consent?

The children or young person's themselves
One parent

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The guardian or primary care giver

Q2.2.P2.2 Will the research involve the participation of any children who are not of sufficient maturity to consent?

Yes

Q2.2.P2.3 Explain how the children or young people's capacity to consent will be judged.

For all children, verbal assent will be sought from individual children at the beginning of the experiment by the child showing assent in their behaviour (e.g., gesture, willingness to follow the examiner to the room) or verbally if they have verbal ability. In addition, a non-verbal indication of willingness to complete the task (e.g., completion of tasks) will be taken as child assent during the assessment session.

Q2.2.P2.4 Describe the form of proposed discussions with children about the research at their level of comprehension.

Considering the age of participants (3-5 years old), it is expected that the children will not be able to fully comprehend the nature of the research. Children will therefore be informed of the nature of the tasks (e.g., looking at pictures, listening to a story) and asked if they are happy to help. If children do not respond or otherwise indicate they do not wish to participate, the session will be terminated.

Q2.2.P2.5 Are you proposing to obtain consent using standing parental consent for the participation of the children or young people?

Yes

(People with a cognitive impairment, intellectual disability or mental illness specific question)

Q2.2.P4.1 Describe the process that you will use to determine a person's capacity to provide consent to participate in the research if that person has a cognitive impairment, intellectual disability or mental illness.

It will be considered consent if the child shows assent in their behaviour (e.g., gesture, willingness to follow the examiner to the room) or verbally if they have verbal ability. Participation will also be considered as consent throughout the experiment. If children do not respond or otherwise indicate they do not wish to participate, the session will be terminated.

Q2.2.P4.2 Describe how the process of obtaining consent is consistent with the requirements outlined in [National Statement 4.5.5-11](#) regarding having respect for participants with a cognitive impairment, an intellectual disability, or a mental illness.

Consistent with National Statement 4.5.5, the decision to participate will be the responsibility of a parent/primary caregiver. Informed consent will be obtained from parents/caregivers prior to any assessments being conducted. Parents/caregivers will be provided with an information package which will provide details of the study and the assessments/procedure their child will undergo if they choose to participate. This information sheet will outline that participation is entirely voluntary and no consequences will be had for non-participation. Consistent with National Statement 4.5.11, the refusal or reluctance to participate in the research by the participant will be respected through termination of the session and their ability to leave at any

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time.

Risk Questions

Q 2.3.1 Describe the risks and burdens associated with your research, referencing any relevant sections of your [Project Description](#) as appropriate.

Autistic individuals: May experience discomfort with the novelty of the robot or situation.
Families/ Parents: Minimal burden is anticipated with parents asked only to read and return the consent form.
AEIOU Staff: Staff will be asked to share previous assessment results of the child and asked to be allowed to use space to conduct the experiment that is within staff presence to maintain ratios (e.g., corner of classroom).

Q 2.3.2 Describe how these risks will be mitigated and managed.

Autistic individuals: Participants are able to stop or leave at any time. Children will also be offered to watch a video of the robot before agreeing to participate. Further, where children are using a visual schedule, this photograph will be placed on their schedule after discussion with room staff/centre managers. We will also provide a social story of the experiment for children prior to engaging in the study.
Families/Parents: We request use of existing assessment results of their child will be used to remove the pressure and burden on families.
AEIOU Staff: Where available (ratio permitting) or required by individual children, conduct of the experiment will occur in an assessment room on location.

Benefit Questions

Q2.4.1 Describe the benefits associated with your research, referencing any relevant sections of your [Project Description](#) as appropriate.

Service provision models/family functioning: This research will provide foundational knowledge towards exploring the use of social robots for learning and intervention with autistic children, particularly for vocabulary expansion. This work will help inform and guide future early interventions aimed at autistic children.
Contribution to current knowledge base: This study will contribute to the literature surrounding autism and child-robot interactions in an educational setting and will further contribute to potential methods of increasing engagement of autistic children in a learning environment.

Q2.4.2 Explain how benefits of this research justify any risks or burdens associated with the research.

Due to the scarce research conducted to date, advancing knowledge and understanding regarding novel word learning for autistic children, will pave the way for the formulation of targeted interventions. This will have long-term benefits for children in terms of improving their

communication and preparedness for school. The future oriented goal of social robots for autistic children is to help support and alleviate burdens on families and service providers while providing social and emotional support for autistic children in ways favourable and comfortable to the child.

Q2.4.3 How will you manage participants' expectations of the perceived benefit of participating in the research?

When completing the consent form, parents and AEIOU staff can select to be emailed a brief description of the findings and they will be presented with a debrief sheet at the conclusion of their participation. In addition, parents will be sent a summary of the results and AEIOU staff will be provided the key findings to distribute to the staff. A copy of the final publication will also be sent to AEIOU's research manager if accepted for publication. Parents or AEIOU staff are invited to contact any of the research team if they have any questions about the project and/or the results.

Section 3 – Data and Privacy

Data Characteristics

Q3.1 Indicate the type of information/data you will be collecting for this project.

Personal information

Q3.2 Indicate the type of information/data you will be using in this project:

Personal information

Q3.3 Indicate the degree of identifiability of information/data you will be collecting for this project.

Individually identifiable information

Q3.4 Indicate the degree of identifiability of information/data you will be using in this project.

Re-identifiable (coded) information

Q3.5 Describe any ethical considerations relating to the collection and/or use of the information/data in this project.

Video recordings of the children will be stored on a secure hard drive and on the [REDACTED] Research Data Management Server. Data will be retained in a deidentifiable form by the [REDACTED]. Participants will be assigned a unique ID number to de-identify their data and data synthesised in a deidentified database. Identifying codes will be stored separately to the database.

Q3.6 Identify the source/s of the information/data that you will be collecting and/or using in this project.

Individual participants and/or relatives or associates of participants

Privately held database

Q3.6.1 Has the data custodian/s, if any, agreed to provide access to the data for use in the proposed research?

Data custodian has approved access to data

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Q3.6.2 Provide any supporting letter or material. (optional)

Decision Letter - [REDACTED] et al. 2023.pdf

Q3.7 Describe any ethical considerations relating to the source of information/data as indicated in the response to the previous question.

Information will be sourced from legal guardians of the children recruited who have given written informed consent. The consent form specifically asks for parent permission to access historical records held by AEIOU for the child's demographic information, Autism Diagnostic Observation Schedule (ADOS) results, Mullen Scales of Early Learning (MSEL) scores, and Vineland Adaptive Behaviour Scales (VABS) scores. Requesting existing data from AEIOU reduces burden on children and parents by avoiding doubling up of assessment.

Q3.8 Was the information/data that you are using previously collected for a purpose other than research?

No

Activities Planned for/with Data

Q3.9 Do you plan to disclose any personal information/data in this project to a third party?

No

Q3.10 How will you protect the privacy of participants and non-participants in any notes and/or publications arising from your research?

Notes taken will be kept securely, either in a locked briefcase for transport, in locked filing cabinets in the school of psychology and then when entered into a database in a password protected online data storage facility ([REDACTED] Research Space) by members of the research team. All participants will be assigned an ID number to de-identify the data collected. All identifying information will be removed from collected data before it is entered into a database, analysed, or published. No potentially identifiable information (i.e. information that could potentially identify individual participants, either alone or in combination such as date of birth and postcode) will be reported in any publications that may arise from these data to maintain individual participant anonymity.

Q3.11 Are there any restrictions on your ability to assure the confidentiality of participants?

No

Q3.12 Do you plan to share any individual research results obtained during this research to the participants?

No

Q3.13 Describe how you will handle any secondary or incidental findings that arise from the analysis of personal information/data.

We do not anticipate obtaining any secondary or incidental findings. However, organisational policies and protocols will be followed depending on the nature of any secondary or incidental findings.

Q3.14 Describe how the information/data will be stored, accessed, archived and/or destroyed.

All identifying information will be removed from collected data before it is entered into a database, analysed or published. No potentially identifiable information (i.e., information that could potentially identify individual participants either alone or in combination such as demographic information) will be reported in any publications that may arise from this data to maintain individual participant anonymity. The interaction of the child with the robot will be recorded using a dedicated device which can only be accessed by the team. Participants will be identifiable in these recordings for coding purposes, but promptly deleted and not published. As such, the data from behavioural coding will be unidentifiable. All data collected and coded will be entered into password protected electronic files, only accessible by [REDACTED] research staff. A file linking the code to the individual's name will be kept in a separate password protected file for the purposes of linking information for compilation of existing data from AEIOU and our data collection. All hard copies of the data will be destroyed once they have been digitised for electronic storage. All data, including audio and/or video recordings will be destroyed after 5 years following completion of the study, subject to agreement between the participants and the researchers. Data kept on file may be used in future research projects that are an extension of, or closely related to, this research. The coding data derived from the video recordings will be identifiable only by a participant number. The data collected will be presented in research publications in a manner that will not be identifiable by a third party. The only other persons with access to this information will be the trained researchers involved in this study.

Q3.15 Describe any ethical considerations relating to the storage of, access to or destruction of information/data in this project.

Recordings will be deleted after coding and analysis have been undertaken. Supplementary data (questionnaires and analysis) will be retained in a secure electronic file at [REDACTED] for a period of five years before being destroyed.

Q3.16 Will the outcomes of this project be disseminated to the participants?

Yes

Q3.16.1.1 Describe how the outcomes of the project will be disseminated to the participants, or refer to the relevant section/s of your Project Description/Protocol which deals with this matter.

When completing the consent form, parents and AEIOU staff can select to be emailed an aggregated summary of the findings. Emails will be stored separately to participant data to preserve anonymity and participants will be provided with a debrief sheet at the conclusion of their participation. In addition, parents will be sent a summary of the results and AEIOU staff will be provided the key findings to distribute to the staff. A copy of the final publication will also be sent to AEIOU's research manager if accepted for publication.

Q3.16.1.2 Describe any ethical considerations relating to any dissemination of outcomes to the participants.

There are no anticipated risks to dissemination of outcomes to participants with only aggregated data shared and no identifying or potentially distressing data to be shared.

Q3.17 Describe any foreseeable future activities for which information/data collected and/or used in this project may be made available.

Information/data collected and/or used in this project may be made available for future research projects that extend upon the current project. This will be limited to de-identified data, results from analyses, and parent information packages. Agreement for the sharing of information will be made between the research team. The data collected will be presented in research publications in a manner that will not be identifiable by a third party. The only persons with access to this information will be the trained researchers involved in this study.

Q3.18 Describe any ethical considerations relating to the planned or possible future use of information/data in this project.

All information/data used in future projects will be de-identified to ensure the privacy of participants. Participants will be notified in the informed consent form regarding future research involving their data and consent obtained.

Section 4 – Attachments and Declarations

Attachments

The following documents have been attached to this HREA.

Project Description/Protocol

See attachment [REDACTED]_S5224905_Honours Proposal.docx

Other attachments

Attachment File Name	Attachment Description
<i>Informed Consent.doc</i>	Informed Consent
<i>Recruitment letter ASD & Robots.pdf</i>	Recruitment Flyer
<i>Debrief Letter ASD & Robots.pdf</i>	Debrief Flyer
<i>Behavioural and Engagement Coding.docx</i>	Behavioural and engagement Coding
<i>Decision Letter - [REDACTED] et al. 2023.pdf</i>	AEIOU Gatekeeper Decision Letter
<i>[REDACTED] Honours Proposal passed letter 2023.pdf</i>	Evidence of prior scientific review. (Optional)
<i>Decision Letter - [REDACTED] et al. 2023.pdf</i>	Supporting letter or material for data custodian's approved access to data. (Optional)

Investigator Team Declarations

The research team has certified that:

- ⌚ All information in this application and supporting documentation is correct and as complete as possible;
- ⌚ I have read and addressed in this application the requirements of the National Statement and any other relevant guidelines;
- ⌚ I have familiarised myself with, considered and addressed in this application any relevant legislation, regulations, research guidelines and organisational policies;
- ⌚ All relevant financial and non-financial interests of the project team have been disclosed; and
- ⌚ In the capacity of a supervisor, as applicable, I have reviewed this application and I will provide appropriate supervision to the student(s) in accordance with the arrangements specified in this application and those associated with the student's educational program.